Fast Weights Recurrent Transformer

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Abstract

The key–value (KV) cache of softmax attention can be viewed as a nonlinear fast-weight network, yet standard sliding-window caching updates it through first-in-first-out (FIFO) replacement. By dropping the requirement for sequence-parallel training, we build on recurrent Transformers and replace FIFO updates with sparse, content-dependent gated refinement of a fixed-size KV memory. On FineWeb, learned refinement reduces validation loss significantly relative to FIFO-based recurrent Transformers at tested memory sizes up to 128 entries. On associative-recall and needle-in-a-haystack tasks, it learns faster and supports successful retrieval in settings where FIFO-based baselines fail, respectively.

1 Introduction

Memory supports ongoing computation by keeping information available until it is needed to make a prediction [Graves et al., 2014]. For example, when you need hot water, remembering that you saw a kettle in the office kitchen helps you decide where to go. This conclusion is not simply recalled from a previous observation, but constructed by combining retained information with a new observation and goal. Memory thus enables information encountered at different times to contribute to computations that the current input alone cannot support [Schlag et al., 2020].

One way to realize such memory in neural networks is through *fast weights*: connection weights that are updated during sequence processing to retain contextual information for later use. In contrast, *slow weights* are learned to encode general knowledge and computational strategies of the entire training corpus [Ba et al., 2016b, Schlag et al., 2020]. Linear-attention models such as DeltaNet instantiates this idea with a fixed-size matrix whose weights encode key–value associations and their readout via matrix-vector multiplication yields a linear query-to-value mapping [Schlag et al., Yang et al., 2024b, 2025b]. The KV cache of standard softmax attention admits a complementary fast-weight interpretation. For each attention head, the cached keys and values can be viewed as the incoming and outgoing weights of a one-hidden-layer network, with a softmax activation between them. Each stored key–value pair corresponds to one hidden unit [Tyulmankov et al., 2021]. This yields a more expressive nonlinear query-to-value map [Cybenko, 1989, Nguyen et al., 2016]. For sliding window attention, this network has a fixed size: each new key–value pair replaces the weights associated with the oldest entry with other entries intact (see Fig. 1b).

Efficient sequence-parallel training places an important constraint on the design of memory updates [Vaswani et al., 2017, Yang et al., 2024b]. In standard attention, each layer computes key–value pairs from its inputs in parallel, and these representations remain unchanged as subsequent positions attend to them. This write-once structure permits parallel attention over a training segment. By contrast, selecting and refining entries based on their evolving contents introduces dependencies between successive memory updates that the standard parallel attention computation can not handle. However, the attention readout itself does not require write-once update. We therefore drop sequence-

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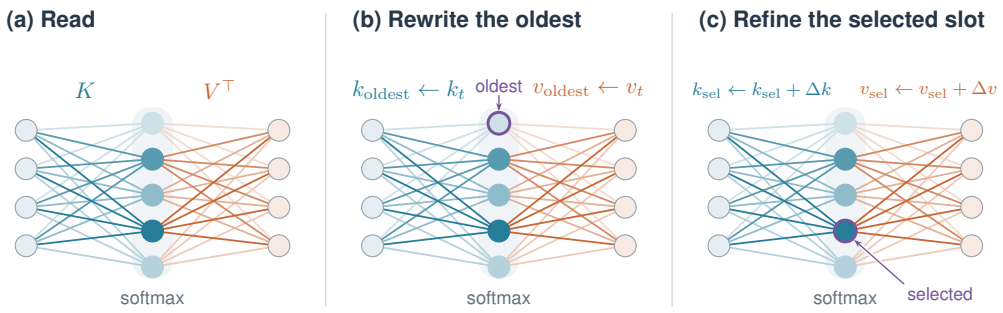


Figure 1: **Recurrent KV memory.** (a) Attention read. (b) Recurrent Transformer's FIFO replacement. (c) FwRT's gated slot revision. One head and a full bank are shown; blue and orange edges denote key and value weights.

parallel training as a design requirement and investigate *whether learned, content-dependent KV cache updates yield a more effective fast-weight memory than FIFO replacement*.

We study this question by building on top of recurrent Transformers [Fan et al., 2020, Oncescu et al., 2026, Cai et al., Anokhin et al., 2026] which demonstrated improved language modeling, and can express computations beyond those available to linear recurrent models [Merrill et al., 2026]. Within this recurrent setting, we retain the attention read operation, but replace the FIFO update rule with learned, content-dependent gated sparse refinement of a fixed-size KV memory (see Fig. 1c). On FineWeb, learned KV refinement reduces validation loss significantly relative to FIFO-based recurrent Transformers at memory sizes up to 128. On associative-recall tasks, it reaches 99% accuracy in fewer training epochs than full-context attention baselines, and on needle-in-a-haystack tasks it supports retrieval at context lengths where FIFO-based baselines fail.

2 Method

We build on the recurrent Transformer (RT) instantiation of the Parallel Experts framework, which evaluates experts in parallel at each depth level while carrying a shared latent state across tokens [Anokhin et al., 2026]. The RT architecture maintains a shared fixed-size KV memory (K, V) across within-layer experts, layers, and tokens. At each depth layer, all experts compute their QKV proposals in parallel from input hidden states h and the current KV memory. The key/value proposals are average-aggregated as $\{k, v\}_{\text{new}}$ and used to update the KV memory in an FIFO manner. The layer's output h_{new} is an averaged readout from the updated KV memory.

$$(q^i, k^i, v^i) = f_{\theta,i}(h, K, V), \qquad i = 1, \ldots, E, \tag{1}$$

$$x_{\text{new}} = E^{-1/2} \sum\nolimits_{i=1}^{E} x^i, \qquad x \in \{k, v\}, \tag{2}$$

$$(K, V) \leftarrow \text{FIFO}((K, V), (k_{\text{new}}, v_{\text{new}})), \tag{3}$$

$$h_{\text{new}} = E^{-1/2} \sum\nolimits_{i=1}^{E} \text{Attention}(q^i, K, V). \tag{4}$$

FwRT. In fast weights recurrent Transformer (FwRT), each expert additionally produces a writing gate vector $\gamma^i \in \mathbb{R}^d_{>0}$. All experts write to the KV memory simultaneously by updating their entries based on normalized dot-product scores between q^i and K.

$$(q^i, k^i, v^i, \gamma^i) = f_{\theta,i}(h, K, V), \qquad i = 1, \ldots, E, \tag{5}$$

$$w^i = \text{softmax}\left(K q^i / \sqrt{d}\right). \tag{6}$$

Colliding writings to the same KV slot is resolved by averaging. The m-th slot of the K and V fast weights are updated using the resolved key/value writings $\Delta\{k, v\}_m$ followed by RMS normalization. The following update rule also applies to the value states.

$$\Delta k_m = |\mathcal{C}_m|^{-1/2} \sum\nolimits_{j \in \mathcal{C}_m} w^j \, \gamma^j \odot k^j, \tag{7}$$

$$K_m \leftarrow \text{RMSNorm}\left(K_m + \Delta k_m\right), \qquad m = 1, \ldots, M \tag{8}$$

where $\mathcal{C}_m$ denotes the expert indices that write to KV slot m. In our experiments, we sparsify w_i so as to access only the top-1 selected slot for each expert, consistent with Fig. 1c.

3 Experiments

We evaluate FwRT on FineWeb [Penedo et al., 2024] language modeling, multi-query associative recall (MQAR), and single-needle-in-a-haystack (S-NIAH) tasks, comparing it with FIFO-based recurrent Transformers and non-recurrent Transformer baselines.

FineWeb. Fig. 2 compares validation loss across memory sizes after 1.31B and 10B training tokens. The non-recurrent baselines use 16 layers, while RT and FwRT use one or two layers with sixteen or eight experts per layer, respectively. The non-recurrent Transformer baselines are trained with sequence length $2M$ and evaluated using a sliding-window KV cache of M entries. At both budgets, FwRT outperforms FIFO-based recurrent Transformers at matched depth for the tested memory sizes up to $M = 128$, but its advantage diminishes as memory grows.

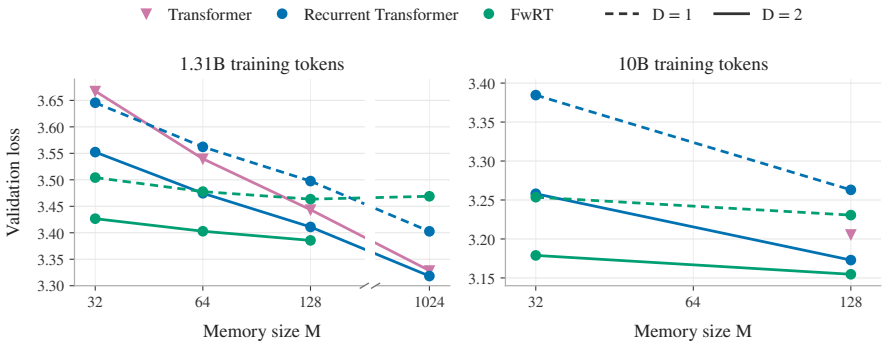


Figure 2: Validation loss vs memory used by a model. D denotes model's depth. FwRT outperforms FIFO-based recurrent Transformers at small memory sizes, but does not scale well with memory size.

Associative recall. Multi-query associative recall (MQAR) [Arora et al., 2024] tests whether models can retrieve values associated with keys encountered earlier in a sequence. We compare learning speed using the first epoch at which test query accuracy exceeds 99%. All models use depth 2 and width 64 consistent with Arora et al. [2024]. FwRT generally reaches this threshold in fewer epochs than our full-context attention and FIFO-based recurrent Transformer baselines (Fig. 3).

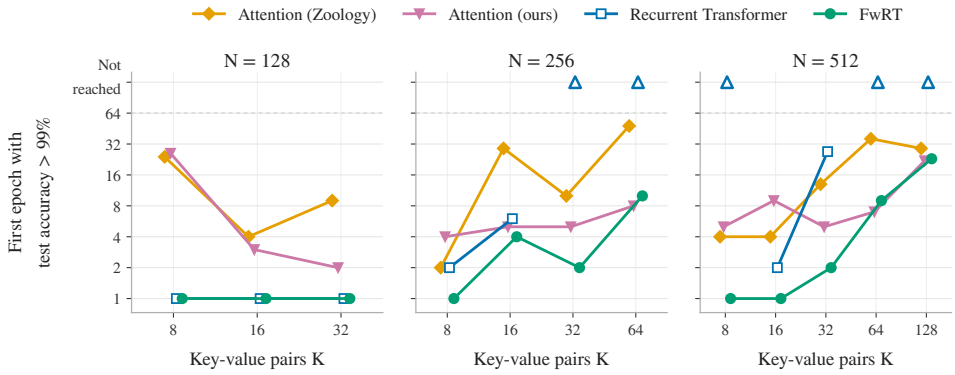


Figure 3: MQAR. Each task pairs sequence length N with K key–value pairs. We report the first epoch above 99% accuracy. Recurrent models use $M = 128$, and attention baselines use full context. Lower is better. FwRT reaches this target accuracy in the same or fewer epochs than the baselines.

Needle-in-a-haystack. We evaluate on RULER's single needle-in-a-haystack suite (S-NIAH) [Hsieh et al., 2024], which hides one key–value pair in a distractor passage and asks for the value given the key, on three tasks: S-NIAH-1, S-NIAH-2, and S-NIAH-3w (a variant of S-NIAH-3 with word-valued rather than UUID needles). All checkpoints are evaluated zero-shot ($n = 500$ examples per cell, greedy decoding) using a re-implementation of `NVIDIA/RULER` with modifications, detailed in Appendix A.

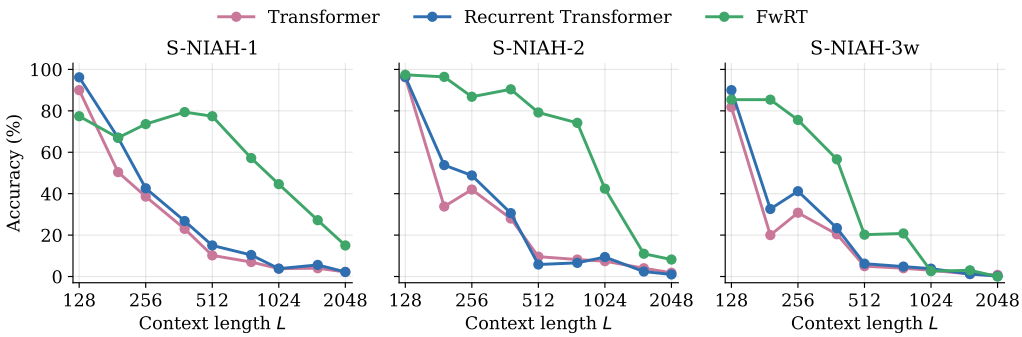


Figure 4: Needle-in-a-haystack accuracy estimated on 500 examples. We evaluate models from Fig. 2 trained on 10B tokens with $M = 128$. FwRT improves needle retrieval accuracy over FIFO-based baselines at context lengths several times its memory size.

Figure 4 shows that FwRT's retrieval advantage over FIFO-based baselines emerges only as context length exceeds the training length, where baselines collapse sharply while FwRT degrades more gradually. All three tasks still converge toward 0% accuracy by $L = 1536$–2048, consistent with a fixed-size, $M = 128$ memory that must eventually be overwritten.

4 Related Work

From Recurrent Transformers to FwRT. Recurrent Transformers [Fan et al., 2020, Oncescu et al., 2026, Cai et al., Anokhin et al., 2026] show that models benefits from recurrence in term of performance. However, the their rigid FIFO strategy of KV cache replacement relies on the temporal order of input data. FwRT proposes to improve it by modeling the KV cache as fast weights, where the slots to update are sparsely selected by the model itself. Raven [Afzal et al., 2026] proposes a similar connection between the FIFO KV cache in Sliding Window Attention (SWA) and persistent memory in State Space Models (SSMs). The Raven architecture is also an instance of a sparse fast-weight model.

Fast weights as memories. Fast weights is proposed in early works by Hinton and Plaut [1987], Schmidhuber [1992] and has been shown to be a strong alternative for modeling memories [Ba et al., 2016a]. Recent progresses of SSMs [Gu and Dao, 2024, Dao and Gu, 2024] and linear attentions [Schlag et al., 2021, Katharopoulos et al., 2020, Yang et al., 2024a, 2025a, Team et al., 2025] popularized the adoption of fast-weight architectures in modern large language models [Team, 2026, Team et al., 2025] since SSMs and linear attentions can be interpreted as fast weights [Behrouz et al., 2025, Wang et al., 2025, Sun et al., 2025]. Sparse update of fast weights have been shown to be an effective approach to scaling up memory size. Mixture-of-Memories [Du et al., 2026] models its fast weights as a mixture-of-experts layer. Alternatively, fast-weight Product Key Memory [Zhao and Jones, 2026] and Sparse Delta Memory [Cabannes et al., 2026] exploits the sparsity of Product Key Memory [Lample et al., 2019].

Recurrent memory. Transformers and SSMs that support parallel scanning trade their expressivity with sequence parallelism [Merrill et al., 2026, Heo et al., 2026]. General recurrent models such as Recurrent Transformers, and FwRT in particular, use a more expressive architectures that carries state across both token positions and model layers.

5 Conclusion

We introduced FwRT, the recurrent Transformer, which replaces FIFO updates with learned, sparse refinement of a fixed-size KV memory. Our experiments show improved language modeling at small memory sizes and successful retrieval in settings where FIFO-based baselines fail. However, FwRT does not scale well with memory size: its language-modeling advantage over FIFO diminishes as the memory bank grows. These results highlight the value of learned KV refinement under small memory budget, and motivate the further work on extending its benefits to larger memory size.

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A Implementation Details and Hyperparameters Used in Experiments

FineWeb. This section reports the training hyperparameters used in the FineWeb experiments. Additional FwRT settings follow the shared and baseline architecture settings. In the learning-rate formulas, d denotes the embedding width.

Table 1: Hyperparameters used in next-token prediction experiments.

Parameter	Value
FineWeb next-token prediction	
Vocabulary size	50,304
Attention heads	8 query, 8 key-value
Token batch size	262,144
Training steps	5,000 or 38,147
Optimizer	Muon core; AdamW embed/readout
Gate parameters	Muon
Optimizer betas	(0.8, 0.95)
Gradient clipping	1.0
Learning rate schedule	Warmup-cosine decay
Peak learning rate	2×10^{-2}
Minimum learning rate	2×10^{-4}
Warmup steps	100
Embedding learning rate	$0.2(d/768)^{-1/2}$
Readout learning rate	$0.004(d/768)^{-1/2}$
Weight decay	0
Cautious weight decay	0.01
Numerical precision	bfloat16
Transformer	
Training sequence length	64, 128, 256, or 2048
Recurrent Transformer	
Shared KV cache length	32, 64, 128, or 1024
Training sequence length	128
FwRT	
Shared KV cache length	32, 64, 128, or 1024
Training sequence length	128
Write-gate and threshold parameters	AdamW
Write-gate learning rate multiplier	0.1
Initial learned write threshold	0.4

FineWeb streaming evaluation. All models are evaluated on the same 100-million-token held-out shard, partitioned into 2,048 contiguous streams. The Transformer uses a layer-specific FIFO KV cache with $M = \textit{training sequence length}/2$ entries, including the current token. RT and FwRT retain their recurrent state and their trained FIFO or learned-slot update rules, respectively.

MQAR. We use Zoology's generator [Arora et al., 2024] with vocabulary 8,192, power-law query placement, zero-valued filler tokens, 100,000 fixed training examples, and 3,000 test examples; loss and accuracy are query-only. All models use two width-64, single-head, pre-normalized attention-only layers with tied embeddings, without MLPs. Attention baselines use full context; recurrent models use $M = 128$. We train in float32 with AdamW $(\beta_1, \beta_2) = (0.9, 0.999)$, weight decay 0.1, batch 128, and epochwise cosine decay for up to 64 epochs. As in [Arora et al., 2024] we sweep learning rate in $\{10^{-4}, 4.64 \times 10^{-4}, 2.15 \times 10^{-3}, 10^{-2}\}$, select the best per model/task by terminal query accuracy, and report the first epoch above 99%.

RULER. We evaluate on RULER's single needle-in-a-haystack suite [Hsieh et al., 2024] at nine prompt lengths $L \in \{128, 192, 256, 384, 512, 768, 1024, 1536, 2048\}$, with $n = 500$ examples per cell and greedy decoding. Every checkpoint is trained once at sequence length 128–256 and evaluated zero-shot beyond it; nothing is fine-tuned. S-NIAH-1 hides a 7-digit number in a repeated filler sentence, S-NIAH-2 hides the same number in Paul Graham essays, and S-NIAH-3w hides a word-valued needle in those essays. We define L as prompt tokens *excluding* generation, where RULER includes it, so that $L = 128$ coincides exactly with the memory size M and the trained sequence length. Recurrent models begin a sequence with an empty memory. We fill it with the prompt's own first M tokens, which keeps the memory in the regime it was trained in.

B Compute Resources

All FineWeb next-token prediction runs use four NVIDIA H100 GPUs, with wall-clock training times of approximately 1–2 hours for 1B tokens and less than 24 hours for 10B tokens. Training a single MQAR configuration takes a few hours on one NVIDIA L40S GPU. RULER evaluation takes several minutes on one NVIDIA A800 GPU.

C Limitations

FwRT does not scale well with memory size M: its language-modeling advantage over FIFO diminishes as the memory grows. It also processes tokens sequentially rather than using parallel scanning. We do not provide ablations isolating the causes of poor memory scaling. Such analysis could guide update rules that better exploit larger memory.

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